

Why MariaDB Scores Over MySQL

This article gives readers an insight into MariaDB as well as MySQL. Both are analysed and the differences between them are discussed. There is also a section on why MySQL users should switch to MariaDB.



Today, organisations have a range of database options to choose from—whether commercial or open source. There was a time when most database management systems (DBMSs) were commercial. But now, with a number of open source databases available, industry professionals prefer to use them. In fact, open source databases have gained a significant share of the market today, because users have the freedom to implement these systems as per requirements and get the best out of them.

Open source database systems are available from various vendors like Oracle, Microsoft, Google, Amazon and Rackspace. The history of open source DBMSs is not very old. The first operational open source DBMS was MySQL, which was launched in 1995 and till date, lots of significant improvements have been incorporated in it, which has made it a strong competitor even to commercial database software.

Open source DBMSs are gaining popularity not only because they are feature-rich and low cost, but also due to their operational efficiency. The freedom users enjoy in customising the source code as per individual needs, and with little programming knowledge, is a plus point. Gartner reports that open source DBMSs are growing by 42 per cent each year, and this industry will be worth US\$ 8 billion by 2020. About 80 per cent of the users polled for the report use some sort of open source software, including DBMSs. Various

commercial vendors like Microsoft and IBM are also offering 'Express Editions' of their DBMS software.

The most widely adopted open source DBMSs are MySQL and MariaDB. This article discusses their similarities, differences and relative merits, before going on to suggest why users ought to migrate to MariaDB.

Databases

Databases are specific structures that hold data in the form of tables. The software that enables the flow of data through a table-based structure is called a DBMS. The most widely used DBMS is a relational database management system (RDBMS). A DBMS is a computer software application that interacts with end users, other applications and the database itself to capture data. A general-purpose DBMS allows definition, creation, querying, updation and the administration of databases.

The functions of a database and its data are:

- **Data definition:** Creation, modification and removal of definitions that lay the foundation for data organisation
- **Data updation:** Insertion, modification and deletion of data
- **Retrieval:** Retrieval of data from the entire database by making specific queries
- **Administration:** User creation and modification, data